Cisco Enterprise Network Topology

Project Documentation | Cisco Packet Tracer

Project detail	Configuration
Platform	Cisco Packet Tracer
Router	Cisco 2911
Switches	Three Cisco Catalyst 2960 switches
Network segments	IT, HR, and Server Farm
Services	DHCP, DNS, HTTP, wireless, SSH, and Telnet

This document describes the network design, addressing plan, device configurations, server services, testing procedures, and troubleshooting concepts implemented in the project.

1. Introduction

This project demonstrates the design, implementation, and configuration of a small enterprise network using Cisco Packet Tracer. The network consists of three departments: IT, HR, and Server Farm. They are interconnected through a Cisco 2911 router and three Cisco 2960 switches.

The project implements IPv4 and IPv6 addressing, DHCP, DNS, HTTP, wireless connectivity, and remote device management using SSH and Telnet. The objective is to gain practical knowledge of networking fundamentals, Cisco IOS commands, IP addressing, network services, and troubleshooting.

2. Project Objectives

• Design a small enterprise network.

• Configure Cisco routers and switches.

• Implement IPv4 and IPv6 addressing.

• Configure subnetting for different departments.

• Implement DHCP services and DHCP relay.

• Configure DNS and HTTP servers.

• Establish wireless guest connectivity.

• Configure SSH and Telnet remote access.

• Test connectivity between different networks.

• Understand basic network troubleshooting.

3. Network Infrastructure

3.1 Network Devices

Device	Model / Name	Quantity
Router	Cisco 2911 (R1)	1
Switch	Cisco 2960 IT-Switch	1
Switch	Cisco 2960 HR-Switch	1
Switch	Cisco 2960 SERVER-FARM	1
DHCP Server	DHCP-SERVER	1
DNS Server	DNS-SERVER	1
Web Server	HTTP-SERVER	1
Wireless Access Point	AP-Guest	1
End Devices	PCs and Guest Phone	Multiple

3.2 Network Architecture

The enterprise network is divided into three independent IPv4 subnets: IT through router interface G0/0, HR through G0/1, and Server Farm through G0/2. The Cisco 2911 router provides communication between these networks. The Server Farm hosts DHCP, DNS, and HTTP services.

4. IPv4 Addressing Scheme

Department	Network Address	Subnet Mask	Default Gateway
IT	192.168.1.0/25	255.255.255.128	192.168.1.1
HR	192.168.1.128/26	255.255.255.192	192.168.1.129
Server Farm	192.168.1.192/27	255.255.255.224	192.168.1.193

4.1 Router Interface Addressing

Interface	IPv4 Address	Subnet Mask	IPv6 Address
G0/0	192.168.1.1	255.255.255.128	2001:DB8:ACAD:1::1/64
G0/1	192.168.1.129	255.255.255.192	2001:DB8:ACAD:128::1/64
G0/2	192.168.1.193	255.255.255.224	2001:DB8:ACAD:192::1/64

5. IPv6 Addressing Scheme

Department	IPv6 Network	Router IPv6 Address
IT	2001:DB8:ACAD:1::/64	2001:DB8:ACAD:1::1
HR	2001:DB8:ACAD:128::/64	2001:DB8:ACAD:128::1
Server Farm	2001:DB8:ACAD:192::/64	2001:DB8:ACAD:192::1

IPv6 routing is enabled on the router using:

```
ipv6 unicast-routing
```

Link-local addresses are configured on the router interfaces using:

```
ipv6 address FE80::1 link-local
```

6. Router Configuration

The router is configured with basic security settings and three routed interfaces.

6.1 Basic Router Configuration

```
enable
configure terminal
hostname R1
enable secret class
service password-encryption
banner motd "Authorized access only"
line console 0
 password cisco
 login
 logging synchronous
line vty 0 15
 password cisco
 login
 logging synchronous
```

6.2 IT Department Interface

```
interface GigabitEthernet0/0
 description Connected to IT-Switch G0/1
 ip address 192.168.1.1 255.255.255.128
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 address FE80::1 link-local
 no shutdown
```

6.3 HR Department Interface

```
interface GigabitEthernet0/1
```

```
 description Connected to HR-Switch G0/1
 ip address 192.168.1.129 255.255.255.192
 ipv6 address 2001:DB8:ACAD:128::1/64
 ipv6 address FE80::1 link-local
 no shutdown
```

6.4 Server Farm Interface

```
interface GigabitEthernet0/2
 description Connected to Server-Farm G0/1
 ip address 192.168.1.193 255.255.255.224
 ipv6 address 2001:DB8:ACAD:192::1/64
 ipv6 address FE80::1 link-local
 no shutdown
```

7. Switch Configuration

Three Cisco 2960 switches are configured for the IT, HR, and Server Farm networks.

7.1 IT-Switch Configuration

```
enable
configure terminal
hostname IT-Switch
enable secret class
service password-encryption
banner motd "Authorized access only"
line console 0
 password cisco
 login
 logging synchronous
line vty 0 15
 password cisco
 login
 logging synchronous
interface vlan 1
 description SVI for IT-Switch Management
 ip address 192.168.1.2 255.255.255.128
 no shutdown
exit
ip default-gateway 192.168.1.1
```

7.2 HR-Switch Configuration

```
enable
configure terminal
hostname HR-Switch
enable secret class
service password-encryption
banner motd "Authorized access only"
line console 0
 password cisco
 login
 logging synchronous
line vty 0 15
 password cisco
 login
 logging synchronous
interface vlan 1
 description SVI for HR-Switch Management
 ip address 192.168.1.130 255.255.255.192
 no shutdown
exit
ip default-gateway 192.168.1.129
```

7.3 SERVER-FARM Switch Configuration

```
enable
configure terminal
hostname SERVER-FARM
enable secret class
service password-encryption
banner motd "Authorized access only"
line console 0
 password cisco
 login
 logging synchronous
line vty 0 15
```

```
 password cisco
 login
 logging synchronous
interface vlan 1
 description SVI for Server-Farm Management
 ip address 192.168.1.194 255.255.255.224
 no shutdown
exit
ip default-gateway 192.168.1.193
```

8. End Device IP Configuration

8.1 IT Department

Device	IPv4 Address	IPv6 Address
IT-PC1	192.168.1.3	2001:DB8:ACAD:1::2
IT-PC2	192.168.1.4	2001:DB8:ACAD:1::3

Subnet mask: 255.255.255.128 | Default gateway: 192.168.1.1 | DNS: 192.168.1.222

IPv6 default gateway: FE80::1 | IPv6 DNS: 2001:DB8:ACAD:192::3

8.2 HR Department

Device	IPv4 Address	IPv6 Address
HR-PC1	192.168.1.131	2001:DB8:ACAD:128::2
HR-PC2	192.168.1.132	2001:DB8:ACAD:128::3

Subnet mask: 255.255.255.192 | Default gateway: 192.168.1.129 | DNS: 192.168.1.222

IPv6 default gateway: FE80::1 | IPv6 DNS: 2001:DB8:ACAD:192::3

9. Server Configuration

Server	IPv4 Address	Subnet Mask	Default Gateway
HTTP-SERVER	192.168.1.221	255.255.255.224	192.168.1.193
DNS-SERVER	192.168.1.222	255.255.255.224	192.168.1.193
DHCP-SERVER	192.168.1.220	255.255.255.224	192.168.1.193

9.1 DHCP Server Configuration

The DHCP server automatically assigns IPv4 addresses to clients in the IT network. DHCP is enabled from the server's Services tab.

Parameter	Value
Service	ON
Default Gateway	192.168.1.1
DNS Server	192.168.1.222
Start IP Address	192.168.1.100
Subnet Mask	255.255.255.128
Maximum Users	27

DHCP Relay Configuration

Because the DHCP server is in the Server Farm network and clients are in the IT network, DHCP relay is configured on router interface G0/0.

```
configure terminal
interface GigabitEthernet0/0
 ip helper-address 192.168.1.220
end
```

This forwards DHCP client requests to the remote DHCP server.

9.2 DNS Server Configuration

DNS service is enabled from the DNS server's Services tab. The following record resolves the domain name to the HTTP server:

Domain Name	Destination
techacadipv4.com	192.168.1.221

An additional DNS record is configured for IPv6 using the HTTP server's IPv6 address: 2001:DB8:ACAD:192::2.

9.3 HTTP Server Configuration

HTTP service is enabled on the HTTP server.

IPv4 address: 192.168.1.221 | IPv6 address: 2001:DB8:ACAD:192::2

The default index.html page is customized with the following content:

```
Project1 Using CISCO Packet Tracer

Welcome to TechAcad Project.
```

Users can access the web page using the HTTP server's IPv4 or IPv6 address through a web browser.

10. Wireless Network Configuration

Parameter	Configuration
Device	AP-Guest
SSID	Guest
Security	WPA2-PSK
Password	Cisco123
IP Configuration	DHCP

The Guest-Phone connects to the wireless network using the configured SSID and WPA2-PSK authentication. The phone obtains its IPv4 address through DHCP and uses automatic IPv6 configuration.

11. SSH Configuration

SSH is configured on the router and IT switch to provide secure remote device management.

11.1 Router SSH Configuration

```
configure terminal
ip domain-name tachacad.com
crypto key generate rsa
1024
ip ssh version 2
username admin password admin
line vty 0 15
 login local
```

```
 transport input ssh
end
copy running-config startup-config
```

11.2 IT-Switch SSH Configuration

```
configure terminal
ip domain-name techacadhelp.com
crypto key generate rsa
1024
ip ssh version 2
username admin password admin
line vty 0 15
 login local
 transport input all
end
copy running-config startup-config
```

SSH Testing

From IT-PC1:

```
ssh -l admin 192.168.1.1
```

From HR-PC1:

```
ssh -l admin 192.168.1.2
```

Enter the configured local username password when prompted.

12. Telnet Configuration and Testing

Telnet is used to demonstrate remote device management. Example commands:

```
telnet 192.168.1.1
telnet 192.168.1.2
```

Telnet uses the configured VTY authentication. Telnet transmits credentials without encryption; SSH is recommended for secure production network management.

13. Network Testing and Verification

13.1 IPv4 Connectivity

Use the following commands from end devices:

```
ping 192.168.1.1
ping 192.168.1.131
ping 192.168.1.221
```

13.2 IPv6 Connectivity

```
ping 2001:DB8:ACAD:1::1
ping 2001:DB8:ACAD:192::2
```

13.3 DHCP Verification

```
ipconfig
ipconfig /renew
ipconfig /release
```

13.4 Router Verification Commands

```
show ip interface brief
show ipv6 interface brief
show running-config
show ip route
show ip dhcp binding
```

13.5 Switch Verification Commands

```
show ip interface brief
show running-config
show mac address-table
```

14. Troubleshooting

Issue	Troubleshooting Approach
DHCP address not received	Check DHCP service, ip helper-address, and connectivity.
Ping unsuccessful	Verify IP address, subnet mask, gateway, and interfaces.
IPv6 communication failure	Check IPv6 addressing and IPv6 unicast routing.
DNS not resolving	Verify DNS service and resource records.
HTTP page unavailable	Check HTTP service and server connectivity.
SSH connection failure	Verify username, RSA keys, SSH version, and VTY settings.
Wireless connection failure	Verify SSID, WPA2 password, and wireless settings.

15. Learning Outcomes

• Enterprise network design using Cisco Packet Tracer.

• Cisco 2911 router configuration.

• Cisco 2960 switch configuration.

• IPv4 subnetting and IPv6 addressing.

• DHCP server and DHCP relay configuration.

• DNS and HTTP server implementation.

• Wireless network configuration.

• SSH and Telnet remote access.

• Network connectivity testing.

• Basic network troubleshooting.

16. Software and Tools

Cisco Packet Tracer, Cisco IOS Command Line Interface (CLI), IPv4 and IPv6 networking, and DHCP, DNS, and HTTP services.

17. Project Files

• Cisco Packet Tracer topology file (.pkt)

• Network topology screenshots

• Project README.md

• Technical project documentation

18. Conclusion

This project demonstrates the implementation of a small enterprise network using Cisco Packet Tracer. It provides practical experience with router and switch configuration, IPv4 and IPv6 addressing, DHCP relay, DNS, HTTP, wireless connectivity, and remote network administration. The project strengthens understanding of networking fundamentals and provides hands-on practice with Cisco networking technologies.

Disclaimer: This project is developed for educational and practical learning purposes. The passwords used in this project are demonstration credentials and must be replaced with secure credentials in production environments.